ch9 Momentum

Q4

4. Train A initially travels at a speed of 60 m s^{-1} along a straight horizontal railway. Another identical train B travels ahead of A in the same direction on the same railway. Due to mechanical failure, B is only travelling at 20 m s^{-1} (Figure 4.1).

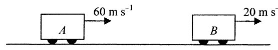
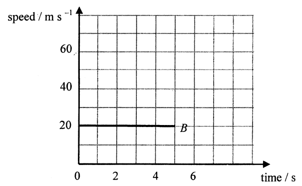


Figure 4.1

At time t = 0, A and B are x m apart, the captain of A receives a stopping signal and immediately A decelerates at 4 m s^{-2} while B continues to travel at 20 m s^{-1}. A eventually collides with B after 5 s. Neglect air resistance.

(a) (i) Find the speed of A just before collision. (2 marks)

(ii) The graph below shows how the speed of B varies with time within this 5 s. Sketch on the same graph the variation of the speed of A within the same period. (1 mark)



(iii) Based on the above information, determine the separation x of the two trains at t = 0. (3 marks)

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(b) A and B locked together after collision.

(i) Find the speed of them just after collision. (2 marks)

(ii) If the collision time between the trains is 0.2 s and the mass of each train is 5000 kg, find the magnitude and direction of the average impact force acted on A during collision. (3 marks)

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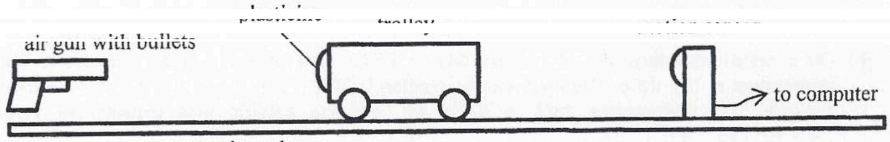
Q2

2. The following experimental items are provided to set up an experiment to estimate the speed of a bullet fired from an air gun.

a smooth track
a trolley
a motion sensor used to measure the speed of the trolley
[illegible]
an air gun and bullets
an electronic balance

The set-up is shown in Figure 2.1.

Figure 2.1



Describe the procedures of the experiment. State the physical quantities to be measured and an equation for finding the speed of the bullet. Write down **ONE** precaution for getting a more accurate result. (5 marks)

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Q4

4. Figures 4.1 (a) and (b) show the cross-section of a fixed spring gun fitted with a small cannon ball.

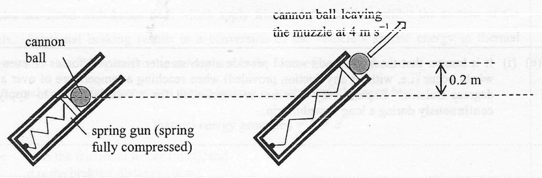

Figure 4.1(a) Figure 4.1(b)

The fully compressed light spring (Figure 4.1(a)) is released so that the cannon ball of mass 0.3 kg leaves the muzzle of the gun at a speed of 4 m s^{-1} (Figure 4.1(b)). Neglect air resistance. ($g = 9.81$ m s^{-2})

(a) During the process from the time when the spring is fully compressed till the cannon ball just leaves the muzzle,

(i) how much energy is transferred from the spring to the cannon ball ? (3 marks)

(ii) explain whether the total momentum of the spring gun and the cannon ball is conserved. (2 marks)

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*(b) The cannon ball is projected at 50° to the horizontal initially and reaches a point horizontally away from the muzzle of the gun at a distance R. Find R and the time of flight t_f of the cannon ball to that point. (4 marks)

*(c) If the projection angle is increased to slightly greater than 50° while the initial speed remains unchanged at 4 m s^{-1}, explain, **without any calculation**, whether there is any change in t_f. (2 marks)

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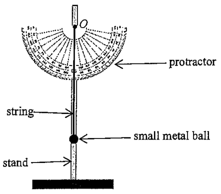
Q10

10. (a) As shown in Figure 10.1, one end of an inextensible string is attached to a small metal ball while its upper end O is fixed to a stand. A protractor is fixed onto the stand behind the string.

Figure 10.1



Write down the steps that you would take to demonstrate the conservation of energy using the set-up given. (4 marks)

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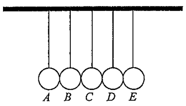
(b) Figure 10.2 shows a Newton's cradle which consists of five identical metal spheres suspended in a row by inextensible light strings of equal length from a fixed horizontal support.

Figure 10.2



Sphere A and B together are displaced slightly to the left and then released from rest. Just after collision, both A and B come to rest while sphere D and E swing together to the right slightly. Neglect air resistance.

(i) (I) Even though the tensions in the strings acting on the spheres are external forces, explain why the law of conservation of momentum can still be applied for this collision. (1 mark)

(II) Suppose that only one sphere (E) swings to the right with speed v_E just after collision. By finding v_E and hence explain why it is not possible for **only one sphere** to swing to the right. Given that the speed of sphere A and B just before collision is 0.50 m s^{-1}. The mass of each sphere is 0.020 kg. (3 marks)

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(ii) The collision repeats alternately from either side for a period of time before the system eventually returns to a rest. Discuss whether the collisions are **perfectly elastic**. (2 marks)

Q5

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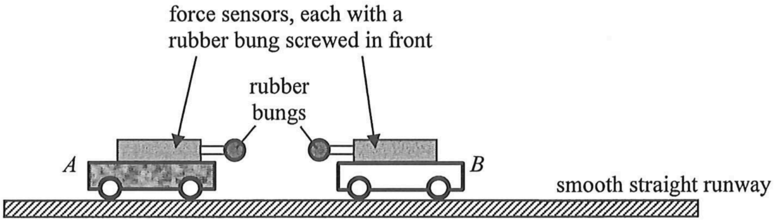
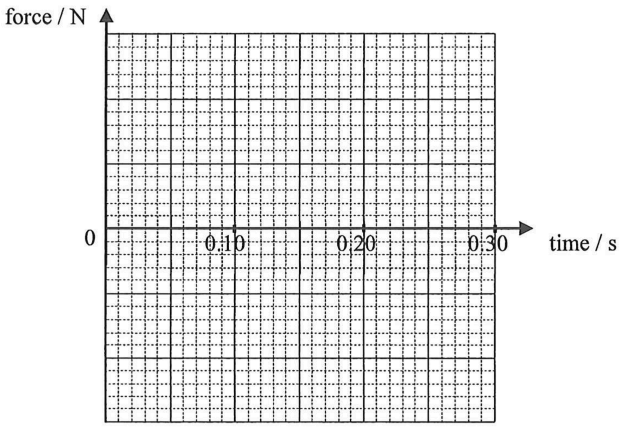
5. You are given two trolleys A and B (with mass of trolley A being larger), which can run freely on a smooth straight horizontal runway. Each trolley has a force sensor mounted on it and the two sensors are connected to a computer interface to record the time variation of force (force towards the right is registered as positive).

Figure 5.1



A student claims that when two objects collide, the less massive object experiences a larger force of collision. Describe how you would conduct an experiment to investigate the forces acting between the trolleys during collision. Sketch the expected force-time graphs of the trolleys in the grid provided and explain briefly. Assume that the collision takes place at time 0 s and lasts for approximately 0.1 s. (4 marks)



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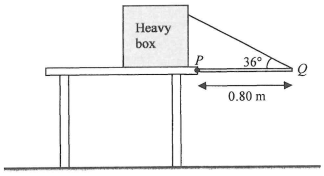
Q3

3. A uniform metal rod PQ of weight 12 N and length 0.80 m is hinged smoothly to the edge of a table at P as shown in Figure 3.1. One end of a light inextensible string is attached to Q and the other end is fixed to a heavy box placed on the table such that the rod is kept horizontal. The string makes 36° with the rod.

Figure 3.1



(a) Find the tension of the string (2 marks)

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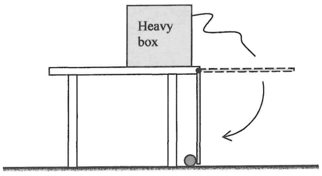
The string is cut. The rod falls, strikes a small ball resting on the floor and exerts a horizontal force on it towards the left, as shown in Figure 3.2. After the collision, the ball moves to the left with speed 1.6 m s^{-1}.

Figure 3.2



(b) (i) If the mass of the ball is 0.05 kg and the collision time is 0.008 s, find the magnitude of the average force acting on the ball by the rod. (2 marks)

(ii) Explain why the momentum of the ball is not conserved before and after the collision. (1 mark)

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